

Active InferAnt Stream 012.2 ~ Case of the Evolving CEREBRUM: The Curious Language of the Incident

ActiveInferAntStream | Case of the Evolving CEREBRUM: The Curious Language of the Incident | 2:16:22

All right.

Hello, welcome.

It is April 16th, 2025.

This is ActiveInference Stream 12.2.

And we will begin it with making a GitHub push to the repo that all of the work is going to be in, continuing on from 12.1.

Start Stream.

This is going to the GitHub repo ActiveInference Institute slash Cerebrum.

And looking forward to people's ideas and questions during this stream.

So this is the case of the evolving Cerebrum.

The curious language of the incident.

The theoretical aspects of Cerebrum.

All right.

So to begin with and sort of catch up on the Zenodo, the version one was uploaded on April 7th.

And commonalities, similar topic.

Also, there was a script that was in an older folder called Cerebrum in the repo that was used to assemble the paper.

In doing that, learned a lot about assembling papers and writing them, expanding them.

And in this current version 1.4, at least today, it's up to 146 pages and still generated by a single simple script.

Which we'll look at in a second.

And how it's expanded.

So there's this sort of paper component.

And the paper's generated from the paper folder in the repo.

Let's look into it in cursor.

First, how the paper is made.

Then we will look at some cool applied outcomes, like making a linear regression neural network and partially observable Markov decision process model.

Declining those into different cases and getting different kinds of behavior from the same model, but of different model types and getting a little bit clearer of a view in a few specific cases, what it means to have a given generative model in a different case.

Okay.

First the paper, then to the source code and deleting things along the way,

adding things as people add anything in the chat.

All right.

First the paper generation.

Oh, and the documents.

The readme in the paper folder describes this process.

Assemble paper is a script of 551 lines

that looks into the component section and assembles them into the output.

So in the output, there is the PDF that's generated,

which concatenates the title page,

makes the tables of content,

including all the main text sections, however many there are,

and then all of the supplements, however many there are.

So it also outputs a pure markdown concatenated version of the assembly process,

and it outputs an HTML version.

So here's a website, rendering, clickable links,

still has the inserted images.

So a lot of flexibility in how,

and also the paper and the images are also vastly improved.

So there's multiple ways to make static documents,

interactive websites,

and markdown files for easy machine reading and formatting as well.

It's generated from the components folders.

There's templates, so you can add more sections.

Then there's three folders, figures, main text, and supplements.

So the figures are the 15 current figures,

and they have their name and title.

There's more general ways to do it,

using LaTeX, Overleaf, all these kinds of things,

so that the numbering is automatic as well.

So this is kind of a fixed numbering system,

but could be generalized too.

So it has, for each figure, a caption,

title, a caption,

and the mermaid markdown code to produce the figures.

Those get saved into the output as direct PNGs.

And also there's a graphical abstract,

which looks a little better in PNG,

which has the figures and the abstracts.

So kind of like scientific poster style.

Then there are the main text

and the supplemental information sections.

The main text is, again,

determined by the numbering of the file,

but these are markdown files

with no mermaid diagrams,

but citing the mermaid diagram for insertion and linking.

So these are markdown files in the main text.

Title page, abstract introduction methods,

case functions and cognitive models,

model workflows as case transformations,

active inference models, and conclusion.

Then there are the supplemental sections.

And if people have ideas for other supplemental sections to add,

we can just add a 13, 14, etc.

I'll open up the website just to get a look at what's in the supplements.

Supplement 1 is a mathematical details and glossary.

Supplement 2 breaks out the novel proposed linguistic cases.

Supplement 3 focuses on different practical applications.

Supplement 4 is a bibliography type section with related work.

5 goes into more category theory details.

6 gives future directions and roadmap.

7 goes into some computational complexity estimates

for different transformations.

8 goes into more detail on active inference.

9 more information on math.

10, algorithmic details.

11, the formal definitions of the core linguistic cases,

the main ones that are focused on in the paper.

And section 12, a case study on writing a research paper

and how in different sections of the paper,

there's different ways that sections are used

and do different things with respect to each other.

And the ideas and the reader.

So that different cases could be applied

to characterize the nature of a model or a process

in that way for those phases.

Just going through this last one,
just as an example of a supplement
and kind of thing that if we wanted to build another supplement,
someone can make a suggestion.

Precision management in research paper writing,
equivalent to learning rate, forgetting rate,
attention, precision, broadly just describing
modulatory strategies for different kinds of up-weightings
and down-weightings.

Message passing occurring amongst author and manuscript,
amongst sections, manuscript and readers and reviewers
with kinds of intent-oriented entities in different cases
or using a similar joint model of communication in different settings.

Motivating why to have cases.

Writing strategies.

Haven't looked into any of these.

Maybe some are useful.

How to make certain kinds of pivots rhetorically
in, I guess, speech or writing
and different ways to interpret
how regimes of attention change
across different sections.

All right.

So that's the paper on Zenodo.

And we can add more.

Just to close off this section
on the paper assembly process,
there are probably other tools
to do this more flexibly.

There's things like DeepWriter
and a lot of other approaches
to doing structured document generation,
building it from scratch.

with cursor is still pretty interesting.

And having plain text,
multiple sections,
possibly doing lumping and splitting,
like have introduction 3.1, 3.2, 3.3,

or just a more flexible way to order the sections.

It makes it really easy to go into a section
and add some kind of feature using cursor.

Like here in the introduction,
just add one relevant paragraph to the end.

structured, smaller forms,
figuring out the right kinds of course graining.

Obviously, just having a dictionary
isn't writing a paper.

So having things broken out to the word level
might be kind of a granular way,
but maybe there's something interesting about that format.

Having a lookup table of the dictionary
and encoding sections and words
in a totally graphical way.

It's just easier to write and version.

Just small updates to the text can be done and reviewed.

So collaborative editing can zoom in
and break out different sections.

It'd be pretty clear.

If there's a flexible ordering method
or just with the file names,
even if someone broke out 4A and 4B
or 4.1, 4.2,

updated the assembly script, etc.

So, okay, that's the paper.

Next, how about the examples of the specific models,
then the documents,

then anything that people have brought up by then.

Bigger picture,

how do we integrate case-based logic?

So nouns in different roles and sentences,
however, treating models as sort of noun entities.

How do we do that in theory and practice?

Different ways to make visuals.

Mermaid's been super useful.

Maybe there's other plain text formats
that are useful too,

but creating good system and process diagrams,
structuring information from prose and tables and lists.

Math with LaTeX.

All right.

Linear regression.

PalmDP neural networks.

So this is in the tests,
in the source, SRC folder,
tests, subfolder.

Let's see if CodeViz will
give a good representation of this.

Hmm.

Okay.

Maybe not today.

In the tests folder,
in the tests folder,
there are several script files
testing each of these different models,
some utilities,
and readme.

The three that are fleshed out at this point are neural network,
linear regression,
active inference,
POMDP.

So in each of these cases,
there's going to be a single model that gets defined,
and then it's going to be deployed in different ways,
leading to visualizable,
and then it's going to be a little bit more.
So let's look at some of the images first.

Okay.

Here's a graphical summary of how for the very same neural network,
whatever kind of technique is being used,
there are different ways that just considering neural network like a noun,
let alone the referent of what it means.

But there are ways that people talk about models.

That's the curious language,
is the what people talk about when they talk about models,

and it's right there in the sentence,
though in a way silence in most cases in English.
There's the neural network does something.
Standard dictionary form nominative.
The researcher trains the neural network.
Like the researcher looks at the neural network.
Accusative case,
recipient of action.
Direct and indirect,
similar with dative,
more of like to give,
rather than to look upon.
Genitive,
outputs of,
one of the cases that's visible in English with the apostrophe S,
like the table's leg,
so it's the outputs of the table,
and these are,
it's a different way to talk about the outputs of something.
Just at the first pass,
it's used in a different way in the sentence.
And then looking from the way it's used in a sentence,
on up through what the real referent is,
it starts to become a little clearer
how these kinds of sentiments that are already expressed
in natural language,
in method sections,
in people's own inner voice,
and conversations around modeling.
Different ways that the words are used,
correlate,
or relate to,
in this linguistic fashion,
to differences in the deployment of the model.
Hence,
possibly,
regularities that can be described,
and also structured in ways that are outside of these.

Okay,
just the last cases here.
Instrumental,
by means of,
the team solves the problem by using,
or with,
a neural network.

Locative,
the patterns exist within,
the neural networks layer.

So,
referring to the model as a location,
where there are spaces,
within,
or being in time,
within the model.

Ablative,
the errors originate from,
the neural network weight.

It's kind of an interesting case.
has some similarity with,
genitive,
in some situations.

And then the vocative,
which is the direct addressment,
like,
calling a different digital assistant,
with a name,
or,
calling the model,
to see if it exists,
like a ping.

Then,
in each of those cases,
the same neural network,
gets,
run,
with these slight differences.

So,
it's,
it's,
it's pretty variable per,
um,
okay.
So,
in the ablative case,
it was discussing the errors of,
the model.
So,
in the accusative,
the model was being trained.
So,
this is just a trace,
as,
would be in some other kind of,
machine learning,
paper,
describing the target of optimization.
Like,
in the method section,
where they say,
we train the neural network to,
dot,
dot,
dot.
Here in the dative,
was the giving of data.
So,
this is just describing,
as a function of input distribution,
what do different activations look like,
in the model.
So,
as it's being given different gifts,
what patterns are arising.
Okay.

same neural network.

Same neural network.

Here,

uh,

in the genitive case,

is describing,

uh,

function outputs,

output distributions,

some animations,

some animations of generating output.

Instrumental.

this might be showing different ways that statisticians do things with the model.

Looking at inputs,

outputs,

activations.

looking at the structure of the model.

Looking at the model's training history for a given curriculum,

as a kind of empirical data.

Looking at kinds of interpretabilities,

looking at the structure of the neural network,

as,

as something that can be done with,

like,

tool engineering.

All right.

Locative.

Describing possible locations inside of,

not,

not super fleshed.

Showing the model in its agentic operation.

Like,

when people say,

the model does this.

The model learns to fit this.

Working as active agent generating predictions,

though those predictions would be the predictions of.

So,
it could be the same reference.
It could be used in,
in,
within one sentence,
A through the ball to B,
and then B through the ball to A.
So,
within one sentence or across sentences,
different nouns are used in different cases.
And,
they're marked differently from a written,
from a speaking perspective in different languages.
So,
similarly,
models as,
slash,
like nouns,
are used in sentences,
in different case settings.
So,
those have different implications for what is viewed,
what is meant,
what kind of capacities would be required from that model.
like,
does the model need to turn on?
Does it need to just give a ping,
like with a vocative?
Or,
does it need to make a prediction?
So,
that's the neural network example.
Now,
there's two more examples.
There's the,
linear regression,
next.
So,

here,
is a,
y equals mx plus b type,
linear regression.

So,
ablative predictions of,
the model.

There were some very interesting diagrams that were arising in this linear case.

So,
here are,
talking about the model as a causal originator.

So,
there's the model,
which causes predictions,
which influence or lead to decisions,
which have these sort of causal consequences.

So,
that's model as a source of,
of causation,
and prediction.

Cross-validation.

Here,
there's an evaluator,
who is doing something such that they,
it is said that they evaluate that model,
so that the model is being,

here,
evaluated to,
well,
residual analysis,
doesn't,
have anything.

Cross-validation,
in the evaluation,
phase.

Here,
the model is,
in the dative case,

the recipient of data.

Like,

I sent,

I gave a present to my friends.

I sent data to the model.

And that is reflected in the batch ingress process,

with,

different sets of data,

with different sets of data,

or,

like,

perfective and imperfective.

Like,

the streaming online real-time data model,

was receiving data,

in the,

continuous,

and then,

the,

perfective.

Like,

the model,

finished receiving,

50 units of data.

Genitive,

uh,
is like apostrophe S.
So,
it's the possessives of the model.
So,
it's the intercept of the model,
the slope of the model,
mean squared error of the model.
All these kinds of statistical,
expressions.
Things that might be in a results section.
Instrumental.
used as a instrument of action.
Used as a tool.
Like,
we hit the nail with the hammer.
We analyze the data with the model.
So,
it's doing different,
again,
variable functional,
uh,
images.
But,
looking at different kinds.
Okay,
so we trained this model best as we could,
whether it was this or that method.
And then,
now we're going to look at the correlation,
or the causal interaction,
between factors 1 and 2.
And then factors 1 and 3.
And then we're going to use that instrumentally.
Say,
oh,
okay,
here it's,

uh,

0.14 correlation.

And,

and we're this confident,

it's this far away from 0.

And here it's,

this correlation.

So,

this is,

these two features are more correlated.

Or,

they have this causal relationship.

So,

doing things with models.

Locative.

Positions within the model.

So,

where's the model located in information spaces?

That's,

that's one.

That's the kind of interpretation of the locative.

And,

and there are multiple,

um,

senses in which a given case is used.

But,

one of them was highlighted in that neural network example,

which was a location in the model.

Like,

on the third floor of the neural network.

And here,

that would still apply.

You still could talk about,

I'm talking about the,

why,

in the equation.

And then here's another view,

which is the location of the model.

Like,
they are at the store.

The model is at,
two,
comma,
three.

Nominative.

This is sort of,
even though the,
the,
uh,
animation doesn't really work.

The nominative,
nominative is when using the standard dictionary type sense of the word,
when the model does something.

Like,
that model,
fit the data.

That,
entity,
did that.

Model,
as actor,
fitting the data.

Model,
fitting,
data.

All the kinds of sentences where the model is,
it could be replaced like with,
a person's name or a doer.

Vocative.

So here,
this has some,
uh,
there's a few funny ways the vocative is used.

This one is,
in the vocative case,
data points directly address the model.

So,
uh,
each point communicates its residual value to the model line.
This direct communication visualizes the conversation between data,
data and models,
similar to how we might address someone by name.
So this is almost at a sort of algorithm as communication level,
thinking about information processes within like,
oh,
negative 2.55 told the slope where it was and the slope told back its residual.
So that kind of,
whereas in the neural network case,
the vocative focused on the addressment by name for like a digital assistant.
Okay.
Here's some other vocative uses,
even in linear regression.
So it's the data and then the,
the model's prediction.
And then is saying,
I,
I'm telling you how far I am under the value.
Another interpretation of algorithm as communication,
where a given variable X is saying,
Hey,
slope multiply me.
So X times 1.97.
And then to the intercept,
the other part of the MX plus B says add 4.98.
And then combine us both,
or we get combined,
or I'm the final answer after combining.
It's sort of like a linguistic approach to describing information flows and algorithms.
Okay.
Okay.
So that's the linear regression model.
And this is all from like test linear regression,
test neural network,
test active inference,

Palm DP.

So here in the,

in the Palm DP case,

for a given model,

it's being described in different cases.

So just to read the sentences,

I'll just say the model instead of the Palm DP.

The model decides which action it takes next.

Nomative.

The programmer evaluates the model's performance.

Accusative.

The environment gives observations to the model.

Dative.

The model's belief about the state guides decision.

Genitive.

We solve the problem using a model framework.

Instrumental.

Different policies exist in the model state space.

Locative.

The new algorithm deviates from the standard model approach.

Ablative.

And,

hey model,

what action should I take next?

Vocative.

So now let's look through those.

Start at the bottom.

So here is a direct addressment,

like the sort of digital assistant,

addressing a model.

And,

and also which kinds of functions are going to come into play.

So this gets towards one of the use cases for having this type of lexical intelligence for modeling is if one expression block or methods block has a bunch of vocatives,

then you know that for whichever agents are addressed,

these are the only commands that you need to spin up.

Whereas if it was some other case,

like it making a prediction or like,

you know,

doing forward inference might be a different computational setup than getting trained,
than getting evaluated.

So in that way,

just you could say like,

what was happening with one actor through a paragraph where they were being referred to in different cases,
you can trace the same model as it's being referred to within and across sentences in perhaps different ways,
and different computational setups and build for that.

Okay.

So here is focusing on the vocative nature of the call and response between the agents and the environment.
In the nominative.

So here's the model in determination of what to do.

For example,

by doing policy inference,

like expected free energy,

policy selection as inference,

on which actions should I select given the states,

and those might invoke policy inference methods.

So simple two by two matrices for the observation transitions.

So this is tracing out the decision making process,

looking at the POMDP as an agent,

and as it's learning and updating its belief.

So what,

and just here,

just an example animation.

What is its belief and action conditional upon observation?

Okay.

In the locative,

here is using again the,

a little bit of a fusion of referring to policies existing in the state space,

which is pointing to the model's state space.

So in that way,

referring to like a location in the model.

However,

the location of the,

of the,

the location of the state space in the model,

may be locational.

So like,
the,
location parameter of the model,
is in,
three comma four.

So here referring to the points,
the,
phase space location.

So all the kinds of theory and empirical aspects of saying like,
the,
the model oscillated around a point attractor,
or the,
the model made its way from thinking it was a cool day to thinking it was a warm day.

Those are paths and passages in phase space.

So those may use verbs of motion or locative in a certain way.

Instrumental.

People saying stuff like,
we solve the problem using the model.

So showing the model.

Looking at different ways in which the model is used to explore questions.

So making predictions,
looking at phenotype,
computational phenotypes,
like the magnitude of belief updating,
or the,

the,
um,
different aspects of the simulated or actual behavior of,
of model being,
being used as a tool,
looking at different statistical traces from the tool,
looking at error,
um,
envelopes.

Genitive is like the,
the,
the genitive AI,
the generative AI,

like the output of GPT,
or the text wrote,
GPT's text,
the text wrote by GPT.
Predictions of the model.
The model's predictions.
Looking at the beliefs of the model.
Looking at the collection of them.
Looking at the predictions of the model.
Looking at the beliefs of the model
in relation to observations of the model.
Or it could be said,
in relation to observations sent to the model.
So it'd be like another way to use observation
and basically convey the same syntax,
which is like,
we all get it.
It's getting data points.
But,
literally,
as it would be with a noun,
even though the two expressions,
the,
as a function of observations of the model,
as a function of the model's observations,
is using model in the genitive,
as having observations that are possessed by it.
Or,
you could focus on the,
the sender of the data,
and say that the data were sent to it,
or the observations were sent,
or,

X sent observations to the model.
So those two features might both be true.
They might be isomorphic.
They might highlight different features of that setting.

But,
that's just an example
of how many ways there are to map
expressions about models,
model linguistics,
to models in,
in theory and practice,
just like there's a ton of stuff
that could be said about a tree,
and just,
the tree grows.
The leaves of the tree.
People looking at the tree.
Like,
all those things that are kind of kicked off by prepositions.
In context.
Alright.
Last ones for the,
uh,
POMDP,
and then to the documents,
and then to any,
uh,
ideas or questions people have.
Here in the dative,
environment giving observations to the POMDP.
Like,
what,
it's direct,
indirect object.
The POMDP gets observations from the environment.
Is having the POMDP doing something,
getting.
And then,
here,
the environment gives observations to the POMDP.
So,
looking at the impact of observations,

which might describe,
very well,
the same graphic,
or,
or related graphic,
as was used.
Just like,
the mass of the leaves,
and,
the,
branches,
mass,
are referring to different levels of analysis of a tree,
and they're using,
a different case,
but,
they still may be,
very related measurements.
And,
more importantly,
we're talking about the same tree.
So,
this is the same POMDP.
Possibly the same,
run,
of the same POMDP.
But,
if you start to think about,
just,
models as they are spoken,
about the kinds of things people say about models,
like,
first,
we designed the model.
Then,
we use the model to generate this.
Then,
we sent training data to the model.

We got the model's errors,
as it's like,
that kind of talking about models aspect,
sometimes refers to the very same model,
in the exact same situation.
other times,
it refers to the same lineage of model,
being used in different ways.
Like,
if it says,
we use the model to sweep across this parameter range,
or parameter range was swept across,
those kinds of things,
but,
then,
maybe only some of those parameter,
parameterized models were taken forward,
and then they were used to generate predictions.
So,
just,
it could be a variant of something that was described,
like,
not exactly the same,
single file.
But,
in these,
simple examples,
it is like,
the exact same generative model.
All right.
So,
that's the,
a bit about the testing suite,
and,
test-driven development,
is very effective,
in,
this kind of,

augmented coding setting.

It can help lock in a lot of functions,

as can,

good use of Git.

Okay.

Now to the doc section,

which,

actually has many,

many,

many documents.

So,

we will definitely not look at all of them.

But,

at the top level of the docs folder,

there's some information about Cerebrum,

just,

how it works,

this live stream file,

different features about Cerebrum.

Maybe these will go into a folder.

And then there's several folders.

So,

starting from the smaller ones,

dialogues,

these are like,

17 different genres,

of sort of,

creative dialogue,

around and about,

some of these different themes.

So,

it's,

and,

and possible to,

to write more.

So,

just to show what that looks like,

just dragging dialogues in,

or,
could have been at dialogues,
write a new dialogue,
that will be,
very,
YouTube,
appropriate,
for all,
the,
topics,
that are,
referred,
to,
in,
paper,
comma,
it is a live stream.

So,
make,
it,
most,
possible,
like,
most,
YouTube,
live streams do.

Okay.
Discussions,
this is a few,
different comments,
that people have,
stated,
related to the work,
and I just,
responded to,
the,
comments,
in,

writing,

in GitHub,

just because it was easier,

to,

express,

and,

and share it.

Examples,

there's just a few examples,

and we can revisit this one,

too.

Just,

this can,

have a growing number of specific,

the,

the linear,

the linear regression,

but it's just documents related to examples.

So,

here,

let's,

let's do a new,

now there's up to three,

parallel chats,

in,

cursor.

So,

let's write examples up for,

the functional,

taking an output.

Maybe,

in the future,

these kinds of drags will be,

less needed.

They,

they,

they help,

get it right on the first one.

Linear regression,
neural network.
In examples,
please,
make three,
markdowns,
comprehensively,
for the three main tests,
in output.
Have each,
doc,
example,
be,
comprehensive,
explain why,
and how it matters,
why,
and how,
cerebrum,
and cases,
come,
into play,
show,
not tell,
all the kinds of,
things,
pre-addressed,
that anyone,
thinking,
seriously,
about this,
would come to wonder,
initially.
Okay,
back in the dialogue space.
So,
like that,
once there's a genre,

within a,
a folder,
even if it's a hyper,
meta genre,
like,
dialogue,
um,
it's possible to just say,
okay,
or add a new dialogue,
with this,
theme.

Okay,
so that's the examples,
happening.

All right,
then there's the,
insects,
folder.

So,
some of this may have,
uh,
images that fail to render,
but let's look at these,
in GitHub.

Okay,
so here's just,
just zooming in,
just a couple layers,
just a couple prompts,
deeper,
into,
all right,
how do we apply cerebrum,
to entomology.

So,
just like we were previously,
talking about,

well,
the linear regression,
used as a tool,
or the predictions of it.

So,
this is kind of,
uh,
classic standard,
entolinguistics.

What do people talk about,
when they talk about,
bugs?

And,
thinking about,
what,
what functions,
the kinds of things,
people do,
and have,
and will,
and would say,
about insects.

Taking a first pass,
at the syntax,
of what they said,
what are those insects,
being said to do,
or have done to them,
or their location,
et cetera,
in that method section,
or in that natural history report.

So,
it's like,
are we talking about,
different functional roles,
different places,
where,

in describing,
that,
function,
that,
natural linguistic case,
would be used.
different speculative insect cases,
mixing the sort of,
hyper practical,
discourse analysis,
uh,
scientific reproducibility,
avenue,
uh,
literature annotation,
um,
hypercharging,
mixing with,
these speculative cases,
like,
talking about the ants,
pheromonal implication.
That's kind of,
an interesting possibility.
Now,
it might be morphologically silent,
in English,
or some other language,
just as other cases are,
in other languages.
So,
it isn't even that,
the words have to be spelled,
any differently,
or pronounced,
any differently.
It,
it could just be,

a situation,
where it's understood,
that the ant,
as a pheromonal,
actor,
or recipient,
is coming into play.
Maybe these are even,
overmeshed,
with other linguistic cases.
different kinds of topics,
that come up in ants,
but,
different aspects of their activity,
um,
that,
kind of give a special,
spin,
or clarity,
or relevance,
to different ways,
to talk about,
an object,
under different sorts,
of constraints.
looking at,
the ways,
in which,
this,
big difference,
amongst,
languages of the world,
the nominative,
accusative,
distinguishing models,
acting,
from models,
being acted upon,

regardless of the action's,
transitivity.

This might map,
to systems,
with clear initiators,
and responders.

So,
the queen lays an egg.

Irrigative,
absolute.

Groups,
models,
performing,
intransitive actions,
e.g.

moving,
with models,
receiving,
transitive actions,
accusative,
distinguishing them,
from models,
performing,
transitive actions,
e.g.

depositing pheromone.

This might map,
to systems,
where the state change,
of a model,
is paramount,
regardless of whether,
it caused,
or received it.

Like,
whoa,
whole,
ontological,

assertional,
frameworks,
where different languages,
are like,
falling in different ways,
across these two different,
kinds of grammar systems.

And,
maybe there are situations,
where one or the other,
has a different feature.

Defined as,
all the ways,
in which they're different,
are ways in which,
they could be different,
in function,
or in fitness,
in a situation.

Okay,
that's the least,
technical of them.

Here is,
applying those kinds,
of case,
to pictures.

Like,
imagine if the case,
to picture,
were a suffix,
or a prefix,
or a subscript,
or a font,
or something like,
steganographic,
hidden in the word,
or it's just,
a reading lens.

But it's like,
the projections,
of the neurons,
send data,
give data,
to the calyx.
And it's just like,
so,
again,
it's not the only way,
to use these parts,
of the brain.
It's not associating them,
with parts of the brain.
It's like saying,
there could be,
method sections,
that use,
these,
regions,
and,
describe these processes,
using this kind,
of linguistics.
Different,
computational,
precision parameters,
and how they,
are in causal,
relationship with,
sort of natural,
ethological,
behavioral,
linguistics,
about what is happening.
Like,
this is a classic,
Professor Deborah Gordon point,

with,
the space between,
seeing,
the nest mate,
on the ground,
and then,
identifying the task,
it's doing.

Like,
it is foraging,
or it's part of foraging.

And then,
going from,
the task identification,
to the identity assertion.

Like,
it is a forager.

But it's like,
a nest mate,
that we see nursing,
is that,
a nurse?

Is it like,
is a,
is a person who,
plants trees,
when they're not planting a tree,
still a planter?

That's more,
classic,
and to linguistics.

More from the,
Wittgenstein side.

So,
different ways,
that,
stigma,
energy is studied.

All,
all these kinds of scenarios,
looking at beehive,
foraging,
looking at the algorithms,
of the different ways,
that information flows,
and processes happen,
in the,
relocation.

Okay,
insect,
here's,
more,
acronyms,
and puns,
for,
more topics.

So,
in the,
in the kind of last section,
as we sort of go beyond,
cerebrum,
probably today,
maybe we can use some,
of these,
or another one.

Insect cognitive architectures.

All right.

I was happy to see,
the harvester ants,
Pogonomerumax barbatus.

So,
different,
insect,
species,
and,
ways that they've been studied,

even though,
many,
or all,
could be studied for all,
but there's just many,
model-ish,
organisms,
that have been studied.
Well studied insects.
Looking at,
how different brain regions,
are described of,
if not playing functional roles of,
using different cases.
Again,
like,
are there more mentions,
in the literature of,
we injected the dye,
into the,
projections of the neurons.
Or,
is the,
just like,
in the different methods,
sentences,
of the statistics paper,
it was like,
describing the linear regression,
or the neural network,
or the POMDP,
as getting things done to it,
or doing things,
or being used with.
All of those,
similarly,
happen for,
descriptions of,

insect brain regions.

Different,

classical,

behavioral types.

Keywords for,

literature search.

Like,

let's do a literature search,

on,

spatial learning,

plus,

hymenoptera.

Pull all those papers.

Download all the papers.

Break them into sentences.

Look at every sentence.

Do entity extraction.

Determine what case,

different entities are in.

So,

this is kind of like,

a formal linguistic,

complement,

to,

statistically driven,

n-gram learning.

Like,

classically,

just taking large,

bodies of text,

and then looking,

okay,

we can look at the word frequency,

that's like the one gram.

We can look at the,

word co-occurrence,

the two gram,

and going on,

and on,
and on,
and that kind of,
leads to,
you know,
one thing after another,
to the attention is all you need,
which has,
some kind of,
n-gram,
like basically the context,
and the attention mechanisms,
with a bit more,
nuance,
but a lot like that.

Okay,
more tables,
just linking up,
different ways,
that different functionalities,
different,
you know,
what,
what cases would we want to transfer,
from how we talk about ants,
in order to be able to say,
those kinds of things,
about swarms of robots.

Okay,
more case studies.

Here,
is,
more of a deep dive,
on insect brain analysis,
anatomy.

So,
looking at the different sections,
of the brain,

in different ways.

Tables,

looking at different adaptations,

more brain mapping,

more connection with species,

behavioral functions,

neurotransmitters,

so it's like,

there's,

again,

going to the literature,

and annotating,

where those kinds of things,

are said,

and then,

that is already,

just sort of the,

forensic cerebrum approach,

it's just a purely,

annotative strategy.

And then also,

it,

implicitly and explicitly,

involves,

sketching out,

these state spaces,

sequences,

even where,

some of those,

combinatorics,

are like,

grammatically invalid,

or prohibited,

for one reason,

or another.

Okay,

not all these work,

this one doesn't work,
different brain region,
different insect,
research,
that can be done.

Okay,
then,
there's two folders,
with a lot of documents.

So,
the first one there,
is languages.

So,
languages,
first,
there's a folder,
computer.

Computer language,
this has,
uh,
uh,
several dozen,
computer languages.

So,
let's look at,
go.

We'll just look at,
one,
computer language,
then mainly,
at natural languages.

Overviews,
the code,
different kinds of,
use cases.

Annotates the code,
with,
different,

case annotations,
on different processes.

So,
you can see all of these,
kinds of,
abbreviations.

So,
different ways,
what's,
one,
what,
like,
just on,
looking at,
even just one line,
of code,
one language,
might say,
the data is sent to,
X,
from Y.

Or,
Y sends the data to,
X.

Or,
just,
data,
X,
Y.

It's,
not to go into this,
on the computer side,
but the,
uh,
human languages,
folder,
is,
also,

very,

rich.

There are,

a variety,

of,

languages.

Uh,

the,

readme is,

pretty much,

updated.

Let's just look at a few.

Esperanto.

In the previous stream,

we looked at how,

the,

Hungarian case system,

has,

onto,

on,

and from,

a surface.

So,

sort of,

perfect for,

different functionalities,

of interfaces,

and,

and boundaries.

Inter-Slavic.

Russian English,

goes into more detail,

just grammatically,

and cognitively,

on,

on some common,

first and second language learning,

combinations of Russian,

and English,
like,
why certain,
failures,
of language,
are,
common,
cognitively,
and grammatically.
Sanskrit,
Tibetan.
So,
Esperanto,
overviewing,
the,
case,
just like,
kind of,
a,
more like a,
grammar textbook.
Like a,
really fast,
summarized,
grammar,
textbook.
Sanskrit,
went into a bit more,
uh,
detail here.
Just several times,
it would,
add a couple hundred lines.
I'd say,
awesome,
add more,
interesting,
funny,

useful,
meaningful,
sections.

So,
different,
case,
um,
information.

Different,
ways,
that,
a model,
like an NLP model,
could be used,
according to those cases.

Grammatical number,
one,
two,
many,
where,
some languages might only have one and many,
but,
you know,
it could be helpful.

Are we talking about a single agent?

Are we talking about the dyadic communication setting?

Or are we talking about the,
you know,
three,
three's a party?

Different syntactic speech,
features.

Features.

Features.

Features.

Features.

this could relate to,
if we have a given model,
and just,
we're going to be using Python,
or subscript,
or special characters,
or prefix,
it's like,
whatever,
um,
morphology setting,
you're in with the model,
there might be different kinds of,
of,
um,
certain things might be easier to express,
or others,
or might have different consequences.
All right,
then just kept on going,
and just say,
add more.
So,
lots of more,
um,
more connection with
temporal and modal systems,
like,
it starts to get into the beyond cerebrum stuff,
that we'll do in a few minutes,
but like,
there are other parts of speech,
and other linguistic techniques,
other than case,
even though cerebrum has focused heavily on case,

in these kind of two streams,
slash early versions of the paper,
but of course,
there's a whole bevy of conditionalities,
and moods,
and all these different kinds of things,
which,
like,
could be used to describe,
even just statistical models.
Like,
while this is happening,
if that happens,
then this will be happening at the same time,
as viewed from this perspective.
different languages have different natural expressivities,
ways of modifying how things look and seem.
Some languages modify certain things for sex,
others don't.
Some modify for number differently.
They,

of course,
of course,
have different heritages,
and,
and connotations,
and,
and different concepts that are sort of already there.
And then also,
they,
they all have different,
and,
and including created contrived semantic systems as well,
they have some commonalities and differences.
So,
for certain kinds of computational semantics,
there may be some very small programs,

or some very secure ways to do this or that.

And then Tibetan.

Let's just see.

Trust systems.

Evidentiality markers.

Evidential systems.

Direct sensory evidence.

General knowledge or inference.

Personal experience or witness.

With suffixes.

Direct,

indirect,

witnessed,

hearsay,

and inference.

Also,

think about legal ontologies,

and,

and different ways that this kind of relates there.

So,

that's the languages folder.

There's all the computer languages,

and then,

I mean,

not every single one,

but that is where all of the computer languages are.

And,

the natural languages.

All right.

Last folder to look at is the speculative design.

So,

just to show,

kind of similarly here.

Okay.

Just to quickly return to the examples.

Let's,

push to GitHub,

go to the examples,

and then,

look at the three new that were written just now.

Palm DP,

linear regression,

neural network.

So,

introduction,

laying out,

what are all these different cases about?

How do they apply to the Palm DP?

What does the demonstration do?

Linear regression,

background on linear regression,

laying out what is different,

and why it matters.

And,

the neural network case,

similarly.

Okay.

In the speculative design,

folder,

we will add one more,

document.

Okay.

Okay.

Dragged in speculative design.

In this folder, write the most vastly layered, comprehensive synopticon of the...

I'm just going to drag in the...

Or I'll just drag in the single markdown.

Output.

Assembled paper.

Write the most...

And...

Do it...

Most...

Absurdly...

And information densely...

In the style...

Of...

Something...

That...

Strangely...

Would make a lot of sense to see...

Visually on a live stream about this topic...

Yet be...

Essentially...

Incoherent in the linear reading.

Conveying all the concepts totally.

Hmm...

The user wants me to create a new file, assembled paper...

We're looking at the chain of thought of Gemini.

So called chain of thought.

They want me to create a new file, assembled paper.md.

Not really, I didn't say that title, but let's go with it.

It should be visually interesting, but incoherent.

Plan.

List the files.

Read the files.

Synthesize the content.

Create the file.

All these different...

Strategies.

How to make it incoherent.

How to make it incoherent.

Linear reading.

But make sense visually.

It should be visually interesting but incoherent.

Plan. List the files. Read the files.

Synthesize the content. Create the file.

How to make it absurdly layered.

All these different strategies.

How to make it incoherent linear reading but make sense visually.

How to convey all the concepts.

File list.

Then it lists out the files in speculative design.

It's like you know.

It's not exactly what I thought I wanted to know or see.

But it's starting to get to a very interesting place.

For each snippet.

Randomly or pseudo-randomly.

Maybe based on file order.

Intersperse headings, bold terms, list items, and short sentences phrase.

Into the main synopticon string.

Use heavy markdown formatting.

Prioritize density and visual chaos over linear coherence.

Oh, brother.

Ensure terms concepts from all files are included.

Okay.

Okay.

Still going.

Still going.

This is just the beginning.

Incorporating elements.

It's like this is doing quite many cycles of thinking, reading, doing.

Lots of tokens being generated here.

But meanwhile, let's let it cook.

And there's a few that are pretty fun here.

So.

So.

Okay.

Okay.

So here's a Daisy Dolly Ripple.

Satire.

Free speech.

Usage.

As they all are.

Where it's a.

Sort of.

Early 1900s.

Detective story.

Story.

And the first part is just story.

Then it goes into this.

Truly delightful sequence.

Of.

Looking at the.

First off.

Speculating some.

Some ancient origins.

Looking at.

The difference.

I'll.

Turn off dark.

Looking at different.

Hypotheses.

Sort of like.

This.

Like.

Mystery Science Theater 3000.

Reading.

Of a Dolly Ripple.

Situation.

Talking about.

How.

In their.

Case.

In their.

Situation.

In their.

Case.

In their.

Cases.

How.

Different.

Processes.

Happens.

Different.

Context.

Shifting.

Different.

Processes.

That.

Different.

Ways.

That.

That.

Then.

It kind of.

Comes around.

At the very end.

To.

Highlight.

The unique.

Values.

Of the different.

Languages.

And then.

I.

Believe.

They.

They.

Call it a night.

And have some.

Hungarian dinner.

Okay.

This was one.

That.

I just said.

Make a new.

Speculative.

Design.

Document.

Go.

Go.

Somewhere.

That's.

That's never been done.

Or something.

You know.

Glitch ontology.

And error based learning.

So.

It's like.

Kind of.

Glitch pop.

Which has like.

A visual.

And a.

Auditory.

So.

It's kind of.

A multimodal.

Glitch.

But it also.

Comes up in programming.

And it's kind of.

A digitized format.

That.

So like.

Different ways.

That glitch.

Based.

Methods.

We can fix.

Some.

That.

New tab.

Glitch ontology.

Fix.

And improve.

All the mermaid.

Diagrams.

This is probably.

The longest.

Um.

Gemini.

Sequence.

Oh.

Okay.

So it ended there.

Continue to.

Refine the.

Synopticon.

And read.

All papers.

And then.

Save the.

Final.

Full file.

Could be.

Slight differences.

In mermaid.

Rendering.

Um.

But.

Cool.

Idea.

Glitch.

Based.

Ontologies.

Like.

Ontologies.

Of.

Errors.

For.

Describing.

Systems.

That.

Have.

Certain.

Relationship.

With.

Error.

Okay.

This is a very long one.

But.

But an extremely.

Worthwhile.

One.

Looking.

At.

William Blake.

So.

Started.

With.

Just this.

Bullet point.

Type.

Stuff.

Now.

Starts.

Getting.

Into.

Blakeian.

Engineering.
Like.
The.
Fourfold.
Vision.
Discussed.
Elsewhere.
With.
Single.
Vision.
Being like.
Leading.
Up.
To.
Metacognitive.
Integrations.
So.
It's kind of like.
This is.
Not.
Dissimilar.
Than a.
Three or four.
Layer.
Hierarchical.
Model.
Like.
Thoughtseeds.
Or.
Sandved.
Smith.
Type.
Model.
And then.
It just shows.
Like.
Okay.
Which model.

In that.

Fourfold.

Vision.

Paradigm.

Going.

Into.

Detail.

Like.

This is.

Kind of.

Like.

The.

Retinal.

Layer.

With.

The.

Convolutional.

Neural.

Networks.

Dropout.

Layers.

That.

Kind of.

Stuff.

Because.

It's.

Like.

The.

Sensory.

So.

It's.

Like.

Multiscale.

Hierarchical.

Architectures.

Possible.

Self-similarity.

Of.

The.
Two.
Three.
And.
Fourfold.
Visions.
In.
Terms.
Of.
Their.
Their.
Capacities.
Going.
Into.
Different.
Kinds.
Of.
Functionalities.
Fractal.
Self-similarity.
With.
Eternity.
In.
An.
Hour.
How.
Would.
That.
Really.
Get.
Implemented.
Um.
Different.
Cosmological.
Parallels.
What.
Would.
Really.

Be.
Different.
With.
A.
Blake.
Inspired.
Cerebrum.
Using.
AI.
Well.
Single.
Objective.
Contrary.
Balancing.
Multi-Objective.
Like.
To.
To.
Target.
The.

Creation.
And.
Maintenance.
Of.
Tension.
Or.
Paradox.
Architecture.
Detailed.
Architecture.
Layers.
So.
It's kind of like.
There's the.
Does this.
Accurately.
Describe.

What.

Tharmus.

Is.

In the.

Blakeian.

Ontology.

It's kind of like.

A separate question.

That's interpreting this.

Just as like.

Structured.

Humanities.

Research.

And just.

Is that.

Merit.

Of the.

Different places.

This box.

Could be pointing.

Or connected to.

Does that.

Align.

And.

And.

Surely.

There's multiple.

Takes.

Even for.

One character.

And then it's like.

What are the.

Functional.

Attributes.

Of this kind of.

Architecture.

So what is that.

Transfer path.

From.

Loose.

Or ambiguous.

Or.

Multiple.

Or even.

Contradictory.

Poetic.

Description.

And.

Basically.

All of these.

Kinds of.

Information.

Processes.

Generation.

Reality.

Layer.

Belua.

Generative.

Layer.

That's the.

Two-fold vision.

Three-fold vision.

The.

Golden.

New.

Creative.

Layer.

And then.

Eden.

Integration.

Layer.

Okay.

There's just.

Too many.

Funny.

Things.

This.

This was a.

Funny.

One.

The.

Eternity.

Sunrise.

Development.

So.

2025.

This.

Very.

Year.

Implement.

The.

Core.

Zoas.

Architecture.

Begin.

Contrary.

Processing.

Research.

So.

Little.

Different.

Spin.

Then.

Predictive.

Processing.

Okay.

Contrary.

Processing.

Research.

Hmm.

But that's.

That's.

That's.

Sort of.

Unity.

Is.

Plural.

At.

Minimum.

Two.

Research.

Agenda.

But then.

That's.

Also.

The.

Paradox.

Well.

But then.

Isn't.

The.

Single.

Objective.

Unity.

Is.

Plural.

At.

Minimum.

Two.

So.

How.

How.

To.

Hold.

The.

One.

And.

Two.

Two.

That's.

Like.

The.

Single.

And.

Two.

Fold.

Vision.

Then.

Over.

The.

Years.

Leading.

To.

2029.

Eternity.

Sunrise.

Self.

Evolving.

Cognitive.

Architecture.

Begin.

Albion.

Integration.

More.

Sections.

Then.

Really.

Nice.

Process.

Diagrams.

Super.

Information.

Dense.

Looking.

At.

At.

Different.

Being.

Being.

Able.

To.
Do.
Programmatic.
Generation.
Archetypal.
Description.
Of.
Logistics.
For.
Perception.
Action.
Cycles.
So.
This.
Is.
Kind of.
Like.
A.
Whole.
Cognitive.
Sequence.
It's.
Not.
Just.
Saying.
Well.
There's.
A.
For.
Loop.
That.
Gets.
An.
Observation.
And.
Then.
Puts.
Out.

An.
Action.
It's.
Like.
An.
Observation.
Comes.
In.
This.
Parallel.
Processing.
Like.
Marriage.
Of.
Heaven.
And.
Hell.
Status.
Occurs.
Certain.
Sequences.
Happen.
Then.
There's.

Back.
And.
Forth.
Here.
And.
Then.
This.
All.
Etc.
It could be.
Make.
A.
Dynamic.

Control.

Flow.

In.

The.

Marriage.

Of.

Contraries.

Let's.

Do.

Make.

A.

Vast.

Comprehensive.

Dynamic.

Control.

Flow.

With.

Massive.

Level.

Of.

Detail.

Of.

Process.

Diagrams.

Something.

Completely.

Capturing.

The.

Spirit.

Of.

James.

Joyce.

All.

The.

Most.

Chaotic.

And.

River.

Run.
Esque.
Stuff.
Also.
Do.
The.

Same.
In.
Another.
Design.
File.
For.
Cerebrum.
Plus.
Infinite.
Jest.
Okay.
Wow.
Still.
Plugging.
On.
That.
Those.
Are.
Almost.
Almost.
Certainly.
The.
Two.
Very.
Longest.
Executions.
I've.
Ever.
Seen.
From.
Gemini.

For.
For.
For.
The.
Sheer.
Number.
Of.
Steps.
Okay.
Then.
Last.
One.
Before.
We.
Push.
Those.
Updates.
Look.
At.
The.
New.
On.
This.
Was.
As.
Per.
The.
Request.
Of.
A.
Friend.
On.
John.
D.
Who.
Lived.
From.
1527.

To.

1608.

So.

I.

Was.

Not.

Extremely.

Familiar.

With.

D.

But.

This.

Was.

A.

Really.

Interesting.

Approach.

That.

Brought.

Together.

Those.

Process.

Diagrams.

Different.

Kinds.

Of.

Cartography.

Spatial.

Computing.

Time.

Cryptography.

Secure.

Communications.

So.

It's.

Kind of.

Like.

John D.'s.

Work.

Transferring.

On.

To.

What.

Was.

Said.

In.

That.

Work.

And.

What.

Has.

Been.

Said.

About.

That.

Work.

Onto.

The.

Specification.

Of.

Cryptographic.

Or.

Of.

Like.

Esoteric.

Semantic.

Components.

Correspondence.

Tables.

Active.

Inference.

Topics.

Category.

Theory.

Terms.

Different.

Ciphers.

We can do a whole.

Okay.

That was.

Okay.

That was.

It did the full.

Assembled paper.

Make a.

Fast.

Speculative design.

Document.

On.

Using.

Cerebrum.

Or.

Cryptographic.

Design.

Syntax.

Semantics.

Cognitive.

Nerve.

This one.

Halted early.

Okay.

Let's look at the.

Assembled paper.

at the assembled paper.

Okay, synopticon.

This was prompted again to be incoherent to read out loud, but to give a lot of information when visually looked at.

Okay.

Let's look at those, the Joyce and Infinite Jest, then we'll look at the live chat.

If anyone has an idea or a question, then go into the sort of beyond.

So here it didn't make the file, but that's what's so fun about Cursor.

Put that and improvements and additions into a file with the right name.

And also this chat is getting long, so that's why there's this start a new chat for better results down there.

Let's watch these.

Okay.

Joyce and Jest.

Okay.

Okay.

Infinite Jest.

First one doesn't work.

Bayesian Funhouse Mirrors.

This sounds clean.

It rarely is.

Okay.

And the Joyce.

I don't think I can evaluate the details, but, you know, it's cool.

Okay.

And there's the cryptographic.

That's what, you know, again, it's just, it's the concept.

It's the possibility that these spaces can be defined up front or on the fly, and they can be passed to the right, other agents or people, or just generated from scratch or in the future in a better way.

Okay.

Okay.

So the main piece to sort of leave a little snippet of, again, in this last part, it's going to be just a few what's next in our path.

And then, if anybody has an idea or a question in the live chat, they can write it.

Okay.

So.

Okay.

So.

So.

Let's go into a whole new folder.

Beyond Cerebrum.

Beyond Spec.

Down Cerebrum.

Okay.

Okay.

So, given all that information on Cerebrum and all it can do with grammatical case, let's zoom out and think beyond it.

Let's come up with insects or other inverted-related acronyms

conveying

absolute total scope of what can be considered a, quote,

full linguistic intelligence method,

whereby all the parts of language, verb, noun,

all parts of speech, moods, etc.,

are firstly

formalized into the kind of

operations

which flexibly

can and do

apply to, e.g. linear regression.

Neural network

Bayesian graphical model

then

in that case

with full

formalization

we are only in the

tip of the iceberg

in terms of

Zeno and

Exo linguistics.

Firstly, come to totally

understand that

then

spec out

completely

this, quote,

beyond

Cerebrum effort.

Okay, planning a response.

Pormica.

Hmm.

That's a

that's a nice ring to it.

Myriad

Mantis

Apis

Locust

Vespa

Arachne

Silk

All good.

Okay.

Formica

Locust

Arachne

Let's

go with
Formica.
Okay.
First
idea is
the best one.
continue
just even just pre-prompt
pre-writing what I
believe or expect I will want to say
soon or next
with all needed files for functions
underpinned by same functions described in
core
beyond spec document
which is always growing and improving
iteratively
such as
forensic work
describing
the lexical
environment of a
model in time
in place in use
all modern
machine learning
cutting edge
questions
interpretability
scaling
safety
pure
entomological
linguistic
pragmatism
wow
already jumping to the tools
thought you know

it could be a
it could be a
wow I've never seen a chain of thought
that looked like this
create directory structure
okay
source
a source folder within
then update
the spec
create the
init
scripts
add a readme
okay
now this is the one
this is the part I've never seen
constraint checklist
and confidence
follow cursor rules
yes
here's the cursor rules
one funny prompt in another setting was like
write yourself a prompt
that will lead you to generate the prompt
that will help you generate the greatest
most accurate relevant possible cursor rules file set
okay
so it's like
this is kind of where
cue kits
and catechisms
come into play
which is just
developing
these
sets of questions
so it's like

oh yeah
that's the Q78
or like
oh you
you could
you could just
use the
use the Q600
ask all those
figure it out
you know
okay
placeholder
for lexical environment
so for this one
we can
download
we could scrape
papers
or scrape websites
and then use
anything from
rule-based methods
to
existing software packages
to
LLM
API calls
to
say
okay
given
what we know about
cerebrum
and whatever
this formica
or whatever system is
given that

what are the entities

what cases are they in

yes

given the structure now then

comprehensively

continue

so that all

docs

formal

specs

of the

parenthesis

more than

language

formal type

theory

category

calculus

and

Bayesian

cognitive

functions

are 100%

100%

100%

it's not

sort of

where I

hope to

and will go

over the coming

a little bit

with anyone's

open source

participation

if they're

if they're

interested

listening along
this far
let's read that doc
okay
okay
cerebrum
provides a
powerful framework
for integrating
case-based
reasoning
and
Bayesian
representations
focusing
significantly
on the
role of
grammatical
case
in structuring
meaning
and reasoning
while valuable
case grammar
represents
only one
facet of
the complex
system
that is
human
or potentially
non-human
language
to achieve
a truly
comprehensive

computational
understanding
of linguistic
intelligence
we must
expand our
scope
significantly
new chat
tab
speculative
design
looking
past
even
future
horizons
write
a
vast
comprehensive
speculative
design
doc
with
valid
mermaid
diagrams
and
tabular
intelligence
about
what
glial
maximalism
really
means
parenthesis

in group
discussion
only
okay
okay
define
base
types
for
fundamental
linguistic
units
phony
morpheme
lexical
item
syntactic
constituent
semantic
concept
pragmatic
concept
composite
types
dependent
types
connections
to formal
grammar
frameworks
laying
out
those
types
!
research
challenges
relations

okay
but just
continue
I'll start a new
chat for better
results
continue
comprehensively
with developing
things
have all
needed
documents
in
fractal
docs
folder
have all
source code
and tests
start running
and
developing
the full
simple
modular
test
suite
now
okay
question
from the
chat
what
kind
of
predictions
might

you have
about
case
uses
across
disciplines
experience
in the
field
or
experience
with
the
language
what
would
it
reveal
if
case
use
was
different
across
groups
okay
and
another
question
live
chat
can
we
mechanically
deform
the
cranium
!

to 10x
the
cerebrum!
um
probably
but you
probably
couldn't
get it
back
tucker
but I
think
I think
with a
3d
printer
and a
shipping
container
it could
be
possible
what
would
reveal
if
case
use
was
different
across
groups
there'd
be
so
many
interesting

topics
like
the
discovered
slash
invented
question
of
case
to
what
extent
are
to
on
at
from
of
an
object
fundamental
aspects
of
semantics
are there
other
fundamental
aspects
of
semantics
that
we
get
through
or get
around
in human
language

in other
ways
I
think
that at
the
very
least
a
first
step
not
the
only
first
step
not
not
not
not
the
last
step
but
a
first
step
is
this
kind
of
beyond
cerebrum
approach
where it's
like
first
get the

full
interoperable
category
theoretic
calculus
type
you know
algebra
etc
etc
etc
for
all the
languages
that we
know
about
partially
to
scaffold
and
improve
the
fluency
of
people
and
LLMs
partially
to
understand
what
are
all
the
state
spaces
and

then
if
we
find
that
there's
something
where
there's
always
exactly
three
in
every
language
even
if
they
were
all
the
same
we
could
still
say
like
well
what
if
it
was
four
or
maybe
we

curious

language

of

the

incident

which

is

for

the

incident

which

motivates

linguaging

what

is

the

curious

languages

that

let

that

semantics

get

communicated

okay

some more

great

questions

from

Tucker

we

will

make

documents

okay

okay

it

seems
like
human
language
taking
a
survey
of
different
semantics
and
grammars
around
the
world
doesn't
need
case
and
that
even
semantic
density
is not
specifically
correlated
to
case
versus
no
case
structure
yeah
it's
it's
highly
possible
that

by
kind
of
sandpapering
the
language
of
the
exoteric
markers
of
case
like
English
for the
most
part
it
it
it
could
be
associated
causally
or
just
correlationally
with an
increased
or decreased
information
density
so
consider
different
ways
like
per

phoneme
whereas
like
if
you're
doing
stem
modifying
like
a
Slavic
language
then
maybe
there's
only so
many
modifications
that you
can
overlap
if
you're
doing
suffixing
then
maybe
you can
have
more
but
suffixing
might
lead to
very long
symbol
density
then again

if there's a
certain number
of symbols
that are needed
for the
subscribing
I don't think
there's a
single
answer
especially
in the
absence
of
empirical
evidence
in our
hands
and
mandibles
but
we
could
say
write
a
speculative
design
dissertation
summary
of
all
the
situations
where
human
languages
with

and without
different
case
systems
would
and
would
not
have
different
information
densities
and
designs
considered
at
syntax
semantic
narrative
cognitive
levels
okay
fill in
all
placeholder
methods
start
to
run
log
confirm
the
test
suite
for
formica
okay
glial

maximalism

oh

wow

being a

little bit

recalcitrant

to write

this document

despite

its

absolutely

benign

cellular

nature

write

that

and

improve

that

into

a

file

okay

okay

tucker

wrote

why do

so many

ancient

languages

i don't

know

what

usart

is

usakri

uskari

oh

thank you
for
that
we should
add
this
to
languages
oh
it is
there
as
basc
uskari
why
why do
they have
nutshell
i don't
know
now
fully
write out
that
dissertation
tons
of
tabular
formatted
crystal
good
stuff
parentheses
information
vast
technical
mermaid
diagrams

unique
explanations
and
predictions
unknown
and
under
known
through
linguists
in
2025
wow
still
still
dragging
heels
on the
glial
component
maybe
switch
to
clod
oh
great
question
great
question
let's
ask
perplexity
okay
Tucker
wrote
are
any
modern

languages
in
widespread
use
gaining
in
case
complexity
let's
do the
perplexity
research
base
evolution
today
dot md
okay
that's
interpreting
that
okay
that's
interpreting
that
okay
research
and
fully
empirically
address
this
question
from
my
good
friend
the
answer

must
ring in
his
ears
like
gold
and silver
and he
is a
refined
listener
he
expects
only
the finest
information
okay
meanwhile
perplexity
let's
do
the
slower
or the
faster
mode
of
perplexity
I
don't
know
yeah
that
would
be
this
is
kind

of
a
one
one
approach
that
could
be
used
beast
which
is
Bayesian
evolutionary
analysis
by sampling
tree
let's
write out
this
document
this
is
going
to
be
beast
case
analysis
okay
I'm just
going to
copy in
just one
paragraph
about
beast
oh we

have so
much good
stuff happening
though
okay that
one will
finish soon
that one
will finish
soon
let's just
see where
it got
with the
development
yeah
making good
progress
on
formica
okay
with this
description
of
beast
right up
a fully
technical
nsf
postdoc
proposal
or
using
cerebrum
and
formica
parentheses
lift them

up
if you
need
to do
ancestral
state
bayesian
phenotypic
reconstruction
of
ancient
linguistic
elements
we are
going
way
into
the
past
we
will
use
generative
ai
as
needed
for
creating
creating
creating
fluent
text
audio
and
visual
media
data
given

the
explicit
probabilistic
models
and
empirical
data
we will
be
compiling
and
inferring
on
ancient
and
contemporary
languages
ok
glial
maximalism
oh
it's
ok
it says
it's for
internal
discussion
only
I feel
like I
could share
it here
though
but we'll
reset that
chat
go back
to

gemini
write
it
out
fully
the
full
postdoc
application
thank
you
ok
glial
archetypes
oh
this will
look better
on
get
ok
test
infrastructure
ok
we stop
the agent
after 25
calls
continue
look at
the glial
tests
developing
writing
that
postdoc
application
ok
but

to
Tucker's
question
most
linguists
agree
that
human
languages
tend
to
simplify
their
case
systems
over
time
rather
than
develop
more
complex
ones
of
course
just
like
in
genomic
and
phenotypic
settings
it's like
but then
how was
there a
more
complex

one
the
historical
trend
seen
in
Indo-European
languages
has been
towards
case
reduction
not
expansion
Latin
old
English
old
East
Slavic
and
Sanskrit
all of
them
have
dropped
in
cases
are
we
buying
the
dip
with
respect
to
case
number

I don't
know
maybe
it could
go down
further
not
financial
advice
not
linguistic
advice
okay
counter
evidence
virtually
no
documented
examples
of
modern
widely
used
languages
have developed
more complex
systems

as
Anatoly
Levenchuk
wrote
in
languages
cases
usually
die

out
it's
a
known
process
why
would
they
appear
instead
of
disappearing
right
why
would
they
appear
instead
of
disappearing
absolutely
perplexity
let's
see
there's
no
strong
evidence
that
a
modern
language
but
could
a
language
with
more

cases

gain

more

popularity

are

any

languages

gaining

cases

but

English

is

right

now

rare

but

possible

oh

they

debate

about

Mandarin

Chinese

okay

deeper

research

okay

but

now

you

know

what

I am

going

to

ask

next

if

that
case
reduction
is
such
a
slam
dunk
that
it
is
pervasive
today
give
me
the
most
complete
parentheses
and
esoteric
account
of how
those
complex
many
haste
systems
ever
even
came to
be
okay
glial
cells
okay
so
centering

the role
of the
glia
and then
there's
the
neuronal
processing
layer
but it's
like
we don't
really talk
about those
in glia
science
it's more
about the
kind of
multiple
types
of glia
microglial
criticism
okay so
that's
kind of
like the
feedback
okay
centering
the role
of the
oligodendrocyte
between
neurons
using
formica

in the
sort of
neural
semantics
of glial
okay
the emergence
of elaborate
case
paradigms
and languages
such as
proto-indo-european
pi
sanskrit
and finno-ugric
languages
involve
layered
grammaticalization
processes
and sociolinguistic
factors
the
pi
case
system
so this
is the
one that
is mostly
used in
cerebrum
the system
likely arose
through
spatial
and semantic

differentiation

but i

repeat

myself

analogical

extension

case

markers

originally

tied to

specific

semantic

roles

instrumental

for tools

expanded to

abstract

syntactic

functions

phonological

reinforcement

sound

changes

eroded

distinctions

between

stems

prompting

speakers

to innovate

new markers

to avoid

ambiguity

stable

case

systems

!

oh!

some
languages
did have
expanded
case
complexity
i mean
like
maybe
hungarian
or some
of these
other
ones
the
date
of plural
um
persisted
due to
its
role
in
marking
indirect
objects
yeah
interesting
but a
little
little
but
but that's
exactly
that's
exactly
what
we can

analyze
with
this
postdoc
application
that we're
writing
just like
you could
use
beast
to look
at the
genomic
or
phenotypic

patterns
okay
run the
full
formica
test
suite
develop
and improve
the
formica
tests
methods
visualizations
animations
documentation
okay
okay
vastly
expand
the

nsf
postdoc
application
add
comprehensive
tables
specific
explanations
and
predictions
in
huge
valid
mermaid
diagrams
okay
still
plugging
away
on
formica
okay
we're gonna
get this
postdoc
application
done
read
it on
github
see if
anyone
else
has any
questions
or ideas
in the
live chat

and then

call it

a day

yes

okay

yes

develop

everything

fully

run

and improve

that

test suite

which yields full computations and visualizations.

Okay.

All right.

Just save that.

A few improvements on this kind of stuff with like the need to reapply tools

and better situational awareness for the code base,

reading folder structures, kind of doing better in,

that could be partly in cursor, partly other approaches.

All right.

Speculative design beast.

Okay.

Okay.

Reconstructing ancient linguistic phenotypes using integrated Bayesian modeling and case-based reasoning enhanced by generative AI.

Okay.

Okay.

Okay.

Okay.

Intellectual merit.

Pioneering the use of generative AI.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Intellectual merit.

Pioneering the use of generative AI.

Specific generative AI model name.

E.g.

GPT-4.

Tacotron.

I'm sorry.

It is an end-to-end text-to-speech.

Tacotron.

I'm sorry.

It is an end-to-end text-to-speech.

Tacotron.

I'm sorry.

It is an end-to-end text-to-speech.

Tacotron.

Rotter impacts.

Okay.

Oh.

That's going to cost you.

Foible.

Yes.

Lexicon.

Swadesh lists.

Perfect.

It's like there's just so much to learn.

A lot of it is out there.

A Swadesh list is a list of forms and concepts which all languages without exception have terms for.

Okay.

New file just dropped.

Swadesh.

Swadesh lists.

Yes.

Run the test suite.

And add tests.

That show off and visualize.

Formica.

New chat.

Write a Swadesh list for Cerebrum and Formica as language.

Okay.

Thank you, Tucker.

Then write another file.

Meta.

Swadesh list.

Markdown.

Both.

Both maximum length and accuracy.

Yeah.

Cursor writing these terminal commands, running the PyTest by itself.

Super big speedup.

Super effective.

Okay, Hyper Swadesh.

That could be next.

Morris Swadesh.

He started with a list of 215 meanings.

Falsely introduced as a list of 225 meanings in the paper due to a spelling error.

Yeah.

Hate to see that happen.

Whoa.

Whoa.

Wow.

Great, great topic to have.

Okay, this is the Swadesh.

Minimal terminology.

Who, what, what, where, when.

All could be associated.

Develop.

Further.

Further.

And furthermore.

The Swadesh list.

Include all Swadesh terms.

And.

Add.

Case.

Annotation.

Okay.

Then the meta, meta Swadesh.

Well then.

Well then.

Okay.

Surely.

There has been a kind of misunderstanding.

A.

Meta Swadesh.

List.

List.

List.

Is like.

A.

Swadesh.

List.

For.

Such.

Lists.

A list.

Of.

All.

The.

Criterion.

You'd.

Use.

In the production.

Or inference.

Productions.

Or inference.

Of a.

Swadesh.

List.

Coming to a file near you.

Okay.

Formica is up and running.

Add a visualization.

Script.

For.

Formica.

Ensure that.

Running.

The test suite.

Yields.

Copious.

Accurate.

Visualizations.

And.

Dot.

GIF.

Okay.

You're doing okay with the Swadesh list.

Okay.

Let's look at this.

The.

The.

The first.

The layer one Swadesh list.

The L1.

Yes.

Perfect.

Like.

This is the basis of what we need for Formica.

Which is.

What.

What are the.

What are the.

200.

200.

Or so.

Concepts.

That we need in Formica.

For.

Total.

Ant colony modeling.

What are their parts of speech?

Here.

Nest core.

There.

Forage zone.

So it's like.

Home and away.

Like over here.

Over there.

Many.

Many mandibles.

Other.

Alien scents.

One.

Two.

Three.

Four.

Five.

Yeah.

I guess you don't even need six.

Maybe you call that just like.

Legish or something.

Okay.

Different.

Different.

Um.

Different creatures in the nest.

Okay.

Body parts.

Actions.

Natural.

Colors.

Adjectives.

Directions.

Great.

Okay.

Okay.

Now.

We're on the right page.

With the meta.

Swadesh list.

Now.

Write.

That.

Meta.

Swadesh.

List.

Fully.

200.

Concepts.

Or so.

Please.

Please.

Yeah.

Run the test suite.

Confirm and log where visualizations are made.

Okay.

Okay.

I understand.

You want a significantly more detailed and exhaustive breakdown of criteria, principles, and considerations related to Swadesh lists.

Okay.

Okay.

Okay.

Okay.

Okay.

Our next.

Album.

Dot.

Md.

Okay.

New chat.

Pre-diluvian.

Super Swadesh.

Is the name of our next album.

Write out all the song.

Names and lyrics.

And in callouts.

Describe the bat sound.

Okay.

Md.

Swadesh list.

We're there.

It really did go up to 207.

See also.

External resources on Swadesh lists.

Lexico statistics and glottocrinology.

Write another comprehensive document on glottocrinology.

Okay.

Okay.

Okay.

27 tests.

Looks like beyond cerebrum is coming together.

Okay.

Let's see where the saving things to.

Yes.

Run the script.

And then run a terminal command.

To confirm where the visualization is saved to.

Glottocrinology is a controversial method.

Within historical linguistics.

That aims to estimate the time depth.

Since two or more related languages.

Separate from a common ancestor.

It attempts to do this.

By quantifying the degree of lexical replacement.

Within a specific subset of core vocabulary.

Yeah.

I think Bayesian methods could help a lot there.

Okay.

Our album.

We're going to look at this album on GitHub.

Alright.

This is going to be.

This will be the last piece.

I think.

People get it.

It's possible to.

Go beyond cerebrum.

In like.

A 20 minute period.

Okay.

Okay.

We'll close with this.

The pre-diluvian super swadish.

So it speaks first.

Primordial murmur.

Obsidian shards on the shore.

The cartographer's dream.

That's an instrumental.

Occasional sonar pings.

Or distorted radio static hint at failed surges and forgotten geographies.

Whispers from the ziggurat's core.

Arc construct.

Gopher wood and pitch.

40 days.

Delusion dissolution.

Error at landing.

New soil.

Old sun.

Well.

Hope this has been.

A.

Interesting stream for people.

Uh.

It's all.

Creative commons.

BY.

NC.

ND.

So if you want to.

Work on it.

Feel free to.

Make contributions to the repo.

If you want to collaborate.

On a future.

Part of the Formica stack.

There's.

There's always.

Always a way.

And.

We should be.

We should be on our way.

Oh.

Oh.

Oh.

Okay.

That's it.

Looking forward to.

People's comments.

Thank you.

Glea Maximalist.

And Jeff.

And Tucker.

For the.

Live chats.

Okay.

Bye.

Bye.

Thank you.